Bayesian Uncertainty Quantification for Permeability Inversion in Porous Media

Probability Distributions, Conjugate Priors, Gaussian Random Fields, and pCN-Gibbs Sampling — Full Derivations with Figures

June 20, 2026

Contents

1 Introduction

Estimating the permeability field $k(x)$ of a porous medium from sparse pressure measurements is a canonical inverse problem in subsurface engineering. The forward map is Darcy's law, which is well-posed: given $k(x)$, compute $p(x)$. The inverse direction—recovering $k(x)$ from a handful of noisy pressure readings—is *ill-posed*: many permeability fields can reproduce the same observations within measurement error.

The Bayesian framework converts this into a well-posed problem by placing a probability distribution over $k(x)$ and using data to update it. The result is not a single "best" permeability map but a *posterior distribution* that quantifies how uncertain we remain after the experiment.

This document derives every piece of machinery used in the accompanying Python code, starting from the probability distributions introduced in UQ textbooks (Normal, Gamma, Inverse-Gamma) and building up to the full pCN-Gibbs posterior sampler.

2 The Forward Problem: 1D Single-Phase Darcy Flow

2.1 Governing equation

We consider single-phase, steady, incompressible flow on $[0, 1]$ governed by Darcy's law with no volumetric source:

$$-\frac{\mathrm{d}}{\mathrm{d}x}\left[k(x)\,\frac{\mathrm{d}p}{\mathrm{d}x}\right] = 0, \qquad x \in (0, 1), \tag{1}$$

with Dirichlet boundary conditions $p(0) = p_L$ and $p(1) = p_R$. The permeability $k(x) > 0$ is the *unknown field* we wish to infer.

2.2 Analytic solution

Integrating (1) once gives $k(x)\,p'(x) = C$ (constant flux). A second integration yields:

$$\boxed{p(x) = p_L + (p_R - p_L)\,\frac{R(x)}{R(1)}, \qquad R(x) := \int_0^x \frac{\mathrm{d}s}{k(s)}} \tag{2}$$

where $R(x)$ is the cumulative hydraulic resistance. The Darcy flux (constant throughout the domain) is

$$q = \frac{p_L - p_R}{R(1)}. \tag{3}$$

Equation (2) is computed numerically via the cumulative trapezoidal rule on the N-point grid $\{x_j\}_{j=0}^{N-1}$, giving an exact (to quadrature order) evaluation in $O(N)$ operations.

2.3 Observation operator

Let $\mathbf{x}_s = (x_{s_1}, \ldots, x_{s_m})$ be the sensor locations. The observation operator H extracts the pressure at these points:

$$H : k(\cdot) \mapsto \mathbf{d} = \left[p(x_{s_1}), \ldots, p(x_{s_m})\right]^\top \in \mathbb{R}^m. \tag{4}$$

Composing with the forward solve we write $G = H \circ \mathrm{Darcy}$, so that $\mathbf{d}_{\mathrm{pred}} = G(k)$.

3 Probability Distributions in Bayesian UQ

Every UQ textbook opens with the Normal, Gamma, and Inverse-Gamma distributions. Their relevance is not axiomatic—each one earns its place by solving a specific subproblem.

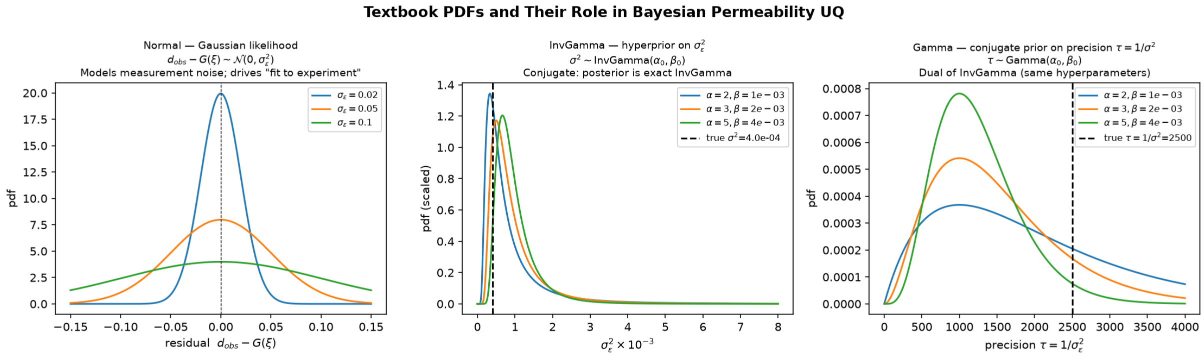

Figure 1: The three foundational distributions for Gaussian-likelihood UQ. *Left*: Normal—models measurement noise and the log-permeability prior. *Centre*: Inverse-Gamma—conjugate hyperprior on the noise variance σ_ε^2. *Right*: Gamma—conjugate prior on the precision $\tau = 1/\sigma_\varepsilon^2$.

3.1 What is a Conjugate Prior? (and Why Should You Care?)

The problem with Bayes' theorem in practice. Bayes' theorem is easy to write down:

$$p(\theta \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \theta)\, p(\theta)}{p(\mathbf{x})}, \qquad p(\mathbf{x}) = \int p(\mathbf{x} \mid \theta)\, p(\theta)\, \mathrm{d}\theta. \tag{5}$$

The numerator is straightforward: multiply the likelihood by the prior. The headache is the denominator $p(\mathbf{x})$, called the *marginal likelihood* or *evidence*. This is a normalising integral over the entire parameter space. For a permeability field on even a modest 100-point grid, θ lives in $\mathbb{R}^{100}$ and this integral has no closed form. In general, computing it requires MCMC, variational inference, or other expensive approximations.

The conjugacy shortcut. A conjugate prior is a choice of prior family that makes the denominator disappear from the calculation.

Definition 3.1. A prior $p(\theta) \in \mathcal{F}$ is **conjugate** to the likelihood $p(\mathbf{x} \mid \theta)$ if the posterior $p(\theta \mid \mathbf{x})$ belongs to the *same parametric family* $\mathcal{F}$, with updated hyperparameters.

When a conjugate prior exists, Bayes' theorem collapses to a handful of arithmetic operations on the hyperparameters. No integrals, no simulation, no approximation — the posterior is exact.

A concrete example: inferring a coin's bias. Flip a coin n times and observe h heads. The Binomial likelihood for bias $\theta \in [0, 1]$ is

$$p(h \mid \theta) = \binom{n}{h}\, \theta^h\, (1-\theta)^{n-h}.$$

We need a prior on θ. The Beta family is conjugate:

$$\theta \sim \mathrm{Beta}(\alpha_0, \beta_0) \qquad \Longrightarrow \qquad p(\theta) \propto \theta^{\alpha_0 - 1}(1-\theta)^{\beta_0 - 1}.$$

The prior and likelihood share the same *kernel shape* — both are products of a power of θ and a power of $(1-\theta)$. Multiplying them (Bayes' numerator):

$$p(\theta \mid h) \;\propto\; \theta^{h+\alpha_0-1}\,(1-\theta)^{(n-h)+\beta_0-1} \;=\; \mathrm{Beta}(\alpha_0 + h,\; \beta_0 + (n-h)).$$

The update rule is just addition:

$$\alpha_n = \alpha_0 + h, \qquad \beta_n = \beta_0 + (n - h). \tag{6}$$

No integral was evaluated. The hyperparameters simply count successes and failures. Crucially, α_0 and β_0 have a natural interpretation as *prior pseudo-counts*: imaginary heads and tails from before any real coin was flipped. Starting with $\alpha_0 = \beta_0 = 1$ (uniform prior) encodes complete ignorance; $\alpha_0 = \beta_0 = 10$ encodes a strong expectation of a fair coin.

Why conjugacy works: kernel matching. The algebraic reason is *shared exponential-family structure*. When the likelihood $p(\mathbf{x} \mid \theta)$, viewed as a function of θ, has the same functional form as the prior, their product stays in the same family. The dreaded normalising constant $p(\mathbf{x})$ is then just the known normalisation of that family, determined uniquely by the updated hyperparameters — so we never need to integrate.

As a formula: if the prior kernel is $h(\theta)^{a_0}\, e^{-b_0\, t(\theta)}$ and the likelihood kernel (in θ) is $h(\theta)^s\, e^{-T\, t(\theta)}$, then the posterior kernel is $h(\theta)^{a_0+s}\, e^{-(b_0+T)\, t(\theta)}$ — the same shape, with parameters updated by simple addition.

What happens without conjugacy. Without conjugacy, the posterior has no closed form and must be approximated. The classic tools are:

- **MCMC** (e.g., Metropolis-Hastings, HMC): generates samples from $p(\theta \mid \mathbf{x})$ without ever computing $p(\mathbf{x})$, but can be slow to mix in high dimensions.
- **Variational inference**: approximates the posterior by the "closest" member of a tractable family (e.g., fully-factorised Gaussian), trading exactness for speed.
- **Laplace approximation**: fits a Gaussian at the posterior mode, valid near the MAP point but misses multi-modality.

Conjugacy avoids all of this.

Relevance to Darcy permeability inversion. Three conjugate pairs appear in our Bayesian sampler, each doing a specific job:

1. **Normal–Normal** (Section 6.1): inferring the mean μ of a Gaussian with known noise σ. The posterior is a precision-weighted average of the prior and the data — data and prior "vote", with weights proportional to how certain each is.

2. **Normal–Inverse-Gamma** (Section 6.2): inferring the variance σ^2 with the mean held fixed. This is *exactly* the update used in the Gibbs step of our pCN-Gibbs sampler. At every MCMC iteration, the noise variance σ_ε^2 is drawn from its exact closed-form posterior — no Metropolis accept/reject step, no tuning, no approximation.

3. **Normal–Normal-Inverse-Gamma** (Section 6.3): inferring μ *and* σ^2 simultaneously. The marginal posterior for μ is a Student-t distribution, automatically heavier-tailed than a Gaussian to reflect the additional uncertainty from not knowing σ^2.

The practical payoff of conjugacy is dramatic. Because the InvGamma update gives a *closed-form sample* for σ_ε^2, what would otherwise require an expensive inner MCMC loop reduces to a single line of arithmetic per outer iteration. This is why UQ textbooks spend so much time on these three distributions before ever mentioning MCMC: conjugacy is the reason the Gibbs sampler is tractable.

3.2 Normal (Gaussian) distribution

Definition 3.2. $X \sim \mathcal{N}(\mu, \sigma^2)$ has pdf

$$p(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The Normal distribution serves three distinct roles in our problem:

1. **Likelihood / measurement noise.** Observations are modelled as $\mathbf{d}_{\text{obs}} = G(k) + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^2 I)$. This is the maximum-entropy distribution consistent with a known variance, and it converts the data misfit $\|\mathbf{d}_{\text{obs}} - G(k)\|^2$ into a log-likelihood.
2. **Prior on the log-permeability field.** Because $k(x) > 0$, we model $Y(x) = \log k(x)$ as a Gaussian random field $Y \sim \mathcal{GP}(\mu_Y, C)$, which guarantees $k = \exp(Y) > 0$ by construction.
3. **KL coefficients.** The Karhunen-Loève expansion (Section 4) parametrises $Y(x)$ through coefficients $\xi_i \sim \mathcal{N}(0,1)$, reducing the infinite-dimensional inference to a problem in $\mathbb{R}^M$.

3.3 Gamma distribution

Definition 3.3. $X \sim \text{Gamma}(\alpha, \beta)$ with rate β has pdf

$$p(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0,$$

with $\mathbb{E}[X] = \alpha/\beta$, $\text{Var}[X] = \alpha/\beta^2$.

The Gamma is the *conjugate prior* for the precision $\tau = 1/\sigma_\varepsilon^2$ of a Gaussian likelihood: if $\tau \sim \text{Gamma}(\alpha_0, \beta_0)$ then, after n observations with sum of squared residuals SS, the posterior is $\tau|\mathbf{d} \sim \text{Gamma}(\alpha_0 + n/2,\ \beta_0 + \text{SS}/2)$.

3.4 Inverse-Gamma distribution

Definition 3.4. $X \sim \text{InvGamma}(\alpha, \beta)$ with scale β has pdf

$$p(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} e^{-\beta/x}, \quad x > 0,$$

with $\mathbb{E}[X] = \beta/(\alpha - 1)$ for $\alpha > 1$ and $\text{Mode}[X] = \beta/(\alpha + 1)$.

If $X \sim \text{InvGamma}(\alpha, \beta)$ then $1/X \sim \text{Gamma}(\alpha, \beta)$, so Gamma and Inverse-Gamma are in a dual relationship. We use InvGamma as the conjugate prior for σ_ε^2 because it is the natural distribution for a variance parameter (positive, right-skewed, supported on $(0, \infty)$).

The key closed-form update is derived in Section 6.2:

$$\sigma_\varepsilon^2 \sim \text{InvGamma}(\alpha_0, \beta_0) \implies \sigma_\varepsilon^2 \,|\, \mathbf{r} \sim \text{InvGamma}\Big(\alpha_0 + \tfrac{n}{2},\ \beta_0 + \tfrac{\|\mathbf{r}\|^2}{2}\Big) \tag{7}$$

where $\mathbf{r} = \mathbf{d}_{\text{obs}} - G(\xi)$ are the residuals.

3.5 Log-Normal distribution and the permeability prior

Because $k = e^Y$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, the permeability itself follows a log-normal:

$$p(k; \mu_Y, \sigma_Y) = \frac{1}{k\,\sigma_Y\sqrt{2\pi}} \exp\left(-\frac{(\log k - \mu_Y)^2}{2\sigma_Y^2}\right), \quad k > 0.$$

The median permeability is e^{μ_Y} and the coefficient of variation grows with σ_Y. Working in log-space (i.e. modelling $Y = \log k$ with a Gaussian field) is both statistically natural and computationally convenient: positivity is guaranteed by construction and the posterior inherits the Gaussian structure of the prior.

4 Gaussian Random Field Prior and KL Expansion

4.1 Gaussian random fields

A Gaussian random field on $[0,1]$ is a stochastic process $\{Y(x) : x \in [0,1]\}$ such that every finite-dimensional marginal is multivariate Gaussian. It is fully specified by its mean function $\mu_Y(x) = \mathbb{E}[Y(x)]$ and covariance function (kernel) $C(x,x') = \mathrm{Cov}[Y(x), Y(x')]$.

We use the exponential (Ornstein-Uhlenbeck) kernel:

$$C(x,x') = \sigma_Y^2 \exp\left(-\frac{|x-x'|}{\ell}\right), \tag{8}$$

where σ_Y^2 is the marginal variance and ℓ is the *correlation length*. Small ℓ produces rough, rapidly-varying fields; large ℓ produces smooth, slowly-varying ones.

4.2 Karhunen-Loève expansion

Let $\{x_j\}_{j=0}^{N-1}$ be the N-point discretisation grid. Define the $N \times N$ covariance matrix $\mathbf{C}$ with $C_{ij} = C(x_i, x_j)$. Its symmetric eigendecomposition gives

$$\mathbf{C} = \mathbf{\Phi}\,\mathrm{diag}(\boldsymbol{\lambda})\,\mathbf{\Phi}^\top, \quad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq 0,$$

where $\boldsymbol{\phi}_i$ is the i-th eigenvector (KL mode).

Retaining only the M leading eigenpairs gives the *truncated KL expansion*:

$$\boxed{Y(x) \approx \mu_Y + \sum_{i=1}^{M} \sqrt{\lambda_i}\,\phi_i(x)\,\xi_i, \qquad \xi_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)} \tag{9}$$

Equation (7) converts the infinite-dimensional field inference into a finite-dimensional problem: *infer the M scalar coefficients* $\xi = (\xi_1, \ldots, \xi_M)^\top$. The fraction of variance captured is

$$f_M = \frac{\sum_{i=1}^{M} \lambda_i}{\sum_{i=1}^{N} \lambda_i}.$$

For the exponential kernel with $\ell = 0.3$ and $N = 65$, the first $M = 20$ modes capture approximately 96.8% of the total field variance.

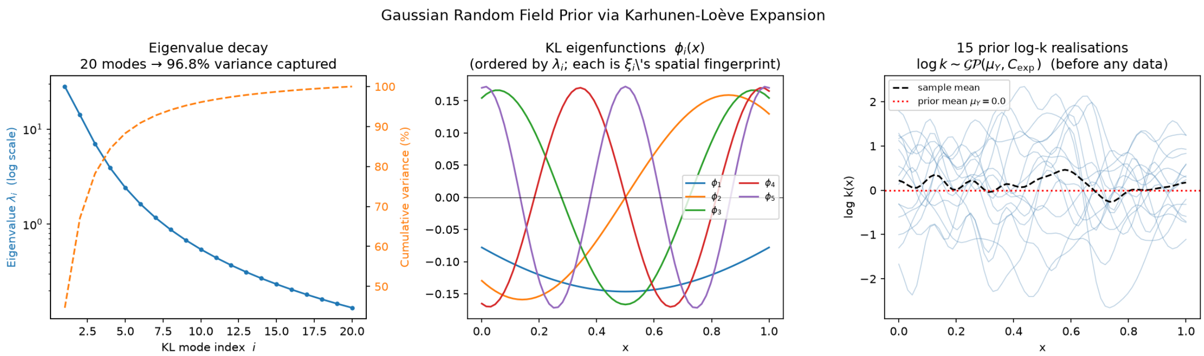

Figure 2: Karhunen-Loève expansion of the log-permeability GRF. *Left*: eigenvalue decay of the exponential kernel; the rapid fall justifies truncation at $M = 20$. *Centre*: first five eigenfunctions (spatial fingerprints of each ξ_i). *Right*: 15 prior sample fields drawn by sampling $\xi \sim \mathcal{N}(\mathbf{0}, I)$ and applying (7).

4.3 Wiener-Askey connection to polynomial chaos

When a surrogate-based approach (polynomial chaos expansion, PCE) is preferred over MCMC, the distribution chosen for the KL coefficients dictates the orthogonal polynomial basis:

Input distribution	Polynomial basis	Support
$\xi_i \sim \mathcal{N}(0,1)$	Hermite	$(-\infty,\infty)$
$\xi_i \sim \text{Gamma}(\alpha,1)$	Laguerre	$(0,\infty)$
$\xi_i \sim \text{Unif}(-1,1)$	Legendre	$[-1,1]$
$\xi_i \sim \text{Beta}(\alpha,\beta)$	Jacobi	$[0,1]$

Since we use $\xi_i \sim \mathcal{N}(0,1)$, the natural PCE basis is Hermite polynomials.

5 Bayesian Formulation of the Inverse Problem

5.1 Bayes' theorem

Let $\xi \in \mathbb{R}^M$ be the KL coefficients (our inferred quantity) and $\sigma_\varepsilon^2 > 0$ the noise variance (treated as unknown, hierarchical model). The observation model is

$$\mathbf{d}_{\text{obs}} = G(\xi) + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^2 I_m), \tag{10}$$

where $G(\xi)$ maps KL coefficients to predicted sensor pressures (via the KL expansion and the Darcy solve).

Bayes' theorem gives the posterior:

$$\underbrace{p(\xi, \sigma_\varepsilon^2 \mid \mathbf{d}_{\text{obs}})}_{\text{posterior}} \propto \underbrace{p(\mathbf{d}_{\text{obs}} \mid \xi, \sigma_\varepsilon^2)}_{\text{likelihood}} \cdot \underbrace{p(\xi)}_{\text{field prior}} \cdot \underbrace{p(\sigma_\varepsilon^2)}_{\text{noise hyperprior}} \tag{11}$$

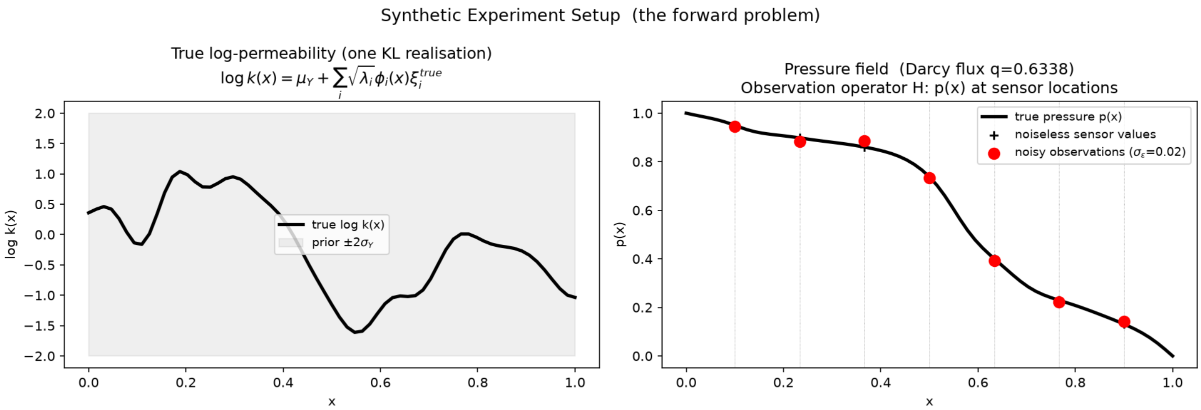

Figure 3: Synthetic experiment setup. *Left*: true log-permeability field (one KL realisation) overlaid on the prior $\pm 2\sigma_Y$ envelope. *Right*: true pressure field and the 7 noisy sensor observations that constitute $\mathbf{d}_{\text{obs}}$.

5.2 Three terms in the posterior

Gaussian likelihood.

$$\log p(\mathbf{d}_{\text{obs}} \mid \xi, \sigma_\varepsilon^2) = -\frac{m}{2}\log(2\pi\sigma_\varepsilon^2) - \frac{\|\mathbf{d}_{\text{obs}} - G(\xi)\|^2}{2\sigma_\varepsilon^2} \tag{12}$$

This is the "comparison to experiment" term: it penalises the squared misfit between the forward model prediction and the observed pressures, weighted by the noise level.

KL-coefficient prior. The KL parametrisation absorbs the GRF covariance into the eigenvectors, so the prior on coefficients is simply:

$$\xi \sim \mathcal{N}(\mathbf{0}, I_M) \implies \log p(\xi) = -\tfrac{1}{2}\,\|\xi\|^2 + \text{const} \tag{13}$$

Inverse-Gamma hyperprior. The noise variance σ_ε^2 is unknown. We place a weakly-informative conjugate prior:

$$\sigma_\varepsilon^2 \sim \text{InvGamma}(\alpha_0, \beta_0), \quad \log p(\sigma_\varepsilon^2) = -(\alpha_0 + 1)\log\sigma_\varepsilon^2 - \frac{\beta_0}{\sigma_\varepsilon^2} + \text{const} \tag{14}$$

5.3 Regularisation interpretation

Dropping the noise hyperprior and fixing σ_ε^2, the log-posterior is

$$\log p(\xi \mid \mathbf{d}_{\text{obs}}) \propto -\frac{\|\mathbf{d}_{\text{obs}} - G(\xi)\|^2}{2\sigma_\varepsilon^2} - \frac{\|\xi\|^2}{2}.$$

Maximising this is exactly *Tikhonov regularisation* with regularisation parameter $\lambda = \sigma_\varepsilon^2$. The Bayesian framework additionally provides: (i) a principled choice of λ via the hyperprior; (ii) credible intervals from the full posterior, not just the MAP point.

6 Conjugate Priors: Theory and Full Derivations

Definition 6.1. A prior $p(\theta \mid \mathbf{h})$ is **conjugate** to the likelihood $p(\mathbf{x} \mid \theta)$ if the posterior $p(\theta \mid \mathbf{x})$ belongs to the same parametric family as the prior, with updated hyperparameters $\mathbf{h} \to \mathbf{h}_n$.

The consequence is that Bayesian updating reduces to a few algebraic formulas—no numerical integration, no MCMC, no approximation.

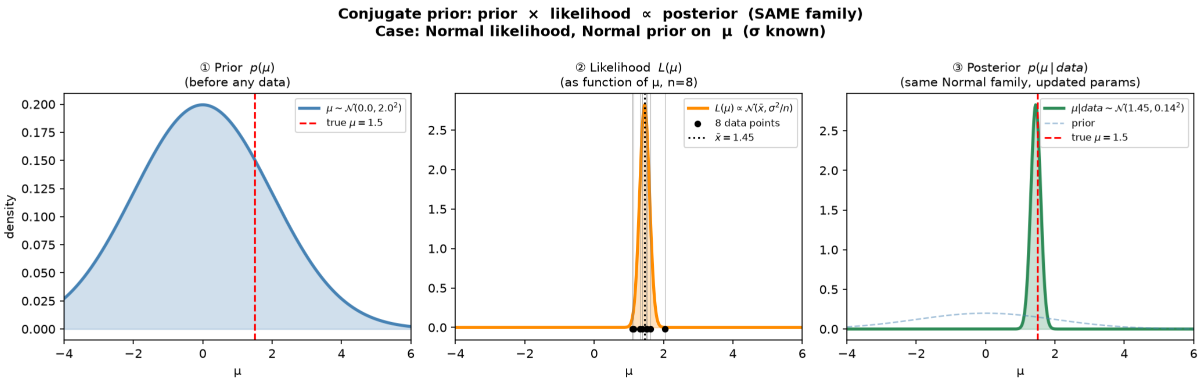

Figure 4: The prior-likelihood-posterior triptych for the Normal-Normal conjugate pair (Case 1). The prior (blue), likelihood function (orange), and posterior (green) are all Gaussians; only the mean and width change.

6.1 Case 1: Normal-Normal (inferring the mean, σ known)

Model.

$$x_i \mid \mu \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2), \quad i = 1, \ldots, n, \qquad \mu \sim \mathcal{N}(\mu_0, \tau_0^2).$$

Derivation. The log-posterior is

$$\log p(\mu \mid \mathbf{x}) \propto \sum_{i=1}^{n} \log \mathcal{N}(x_i; \mu, \sigma^2) + \log \mathcal{N}(\mu; \mu_0, \tau_0^2)$$
$$= -\frac{1}{2}\left[\frac{\sum (x_i - \mu)^2}{\sigma^2} + \frac{(\mu - \mu_0)^2}{\tau_0^2}\right] + \text{const.} \tag{15}$$

Expanding $\sum (x_i - \mu)^2 = n\mu^2 - 2n\bar{x}\mu + \sum x_i^2$ and collecting terms in μ:

$$\log p(\mu \mid \mathbf{x}) \propto -\frac{1}{2}\left[\underbrace{\left(\frac{n}{\sigma^2} + \frac{1}{\tau_0^2}\right)}_{1/\tau_n^2}\mu^2 - 2\underbrace{\left(\frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\tau_0^2}\right)}_{\mu_n/\tau_n^2}\mu\right]. \tag{16}$$

Completing the square gives $\log p(\mu \mid \mathbf{x}) \propto -(\mu - \mu_n)^2/(2\tau_n^2)$, so:

Normal-Normal update

Posterior: $\mu \mid \mathbf{x} \sim \mathcal{N}(\mu_n, \tau_n^2)$ with

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2} \quad \text{(precisions add)} \tag{17}$$

$$\mu_n = \tau_n^2\left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2}\right) = \frac{(1/\tau_0^2)\,\mu_0 + (n/\sigma^2)\,\bar{x}}{1/\tau_0^2 + n/\sigma^2} \quad \text{(precision-weighted mean)} \tag{18}$$

Key insight. Equation (15) shows that the posterior *precision* $1/\tau_n^2$ grows *linearly* in n; equivalently the variance $\tau_n^2 \sim 1/n$ for large n. Adding data is like adding parallel resistors: more resistors $\Rightarrow$ lower total resistance.

Pseudo-data interpretation. The prior $\mathcal{N}(\mu_0, \tau_0^2)$ can be viewed as coming from $n_0 = \sigma^2/\tau_0^2$ imaginary observations all equal to μ_0. Real data updates the same counter: $n_{\text{eff}} = n_0 + n$, $\bar{x}_{\text{eff}} = (n_0\mu_0 + n\bar{x})/n_{\text{eff}}$.

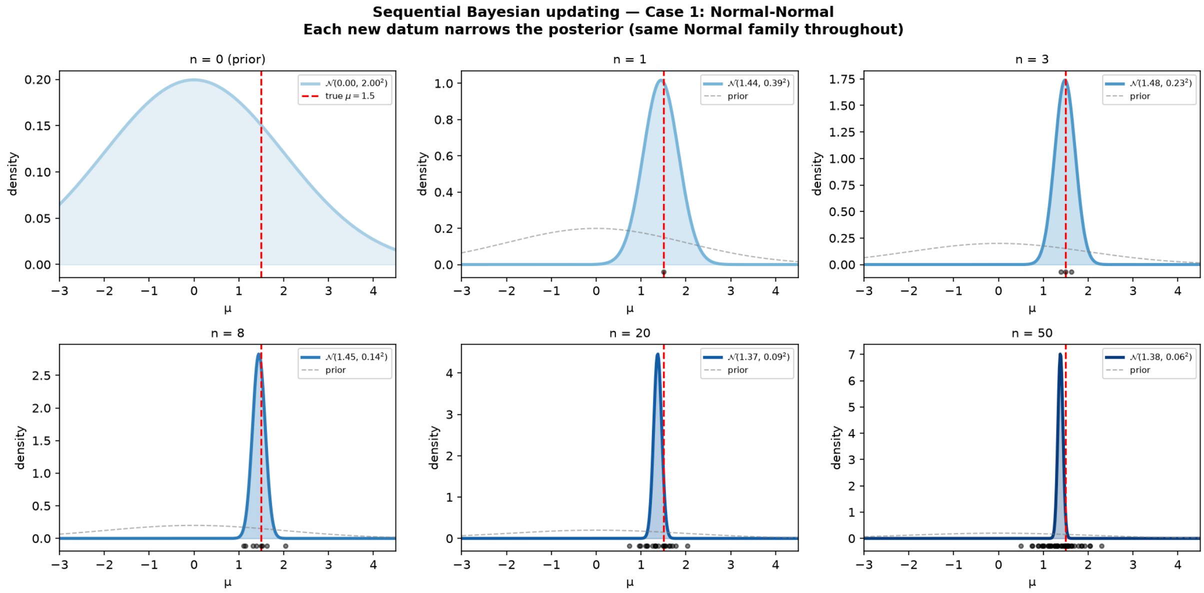

Figure 5: Sequential Normal-Normal updating at $n = 0, 1, 3, 8, 20, 50$. The posterior sharpens around $\mu_{\text{true}} = 1.5$ (red dashed line) as data accumulates. At every step the posterior is exactly Gaussian.

6.2 Case 2: Normal-Inverse-Gamma (inferring σ^2, μ known)

This is the case used directly in the Darcy Gibbs step.

Model.

$$x_i \mid \sigma^2 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2), \quad \sigma^2 \sim \text{InvGamma}(\alpha_0, \beta_0).$$

Derivation. The InvGamma prior pdf is $p(\sigma^2) \propto (\sigma^2)^{-\alpha_0 - 1} e^{-\beta_0/\sigma^2}$. The likelihood is

$$p(\mathbf{x} \mid \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\text{SS}}{2\sigma^2}\right), \quad \text{SS} = \sum_{i=1}^{n} (x_i - \mu)^2.$$

The posterior is therefore

$$\begin{aligned} p(\sigma^2 \mid \mathbf{x}) &\propto (\sigma^2)^{-n/2} \exp\left(-\frac{\text{SS}}{2\sigma^2}\right) \cdot (\sigma^2)^{-\alpha_0 - 1} e^{-\beta_0/\sigma^2} \\ &= (\sigma^2)^{-(\alpha_0 + n/2) - 1} \exp\left(-\frac{\beta_0 + \text{SS}/2}{\sigma^2}\right). \end{aligned} \tag{19}$$

Recognising (17) as an InvGamma kernel:

Normal-InvGamma update

Posterior: $\sigma^2 \mid \mathbf{x} \sim \text{InvGamma}(\alpha_n, \beta_n)$ with

$$\alpha_n = \alpha_0 + \frac{n}{2} \tag{20}$$

$$\beta_n = \beta_0 + \frac{\text{SS}}{2} = \beta_0 + \frac{1}{2} \sum_{i=1}^{n} (x_i - \mu)^2 \tag{21}$$

Posterior mean: $\mathbb{E}[\sigma^2 \mid \mathbf{x}] = \beta_n / (\alpha_n - 1)$ (for $\alpha_n > 1$).

Information accumulation. α_n grows at rate $\frac{1}{2}$ per observation (each datum contributes "half a degree of freedom"), while β_n accumulates the sum of squared residuals. Both updates are additive and can be applied one datum at a time with identical results.

Application to the Darcy Gibbs step. With KL coefficients ξ fixed, the Darcy residuals $\mathbf{r} = \mathbf{d}_{\text{obs}} - G(\xi)$ play the role of $\{x_i - \mu\}$, and $\text{SS} = \|\mathbf{r}\|^2$. Equation (5) gives a *single closed-form sample* for σ_ε^2 without any accept/reject step.

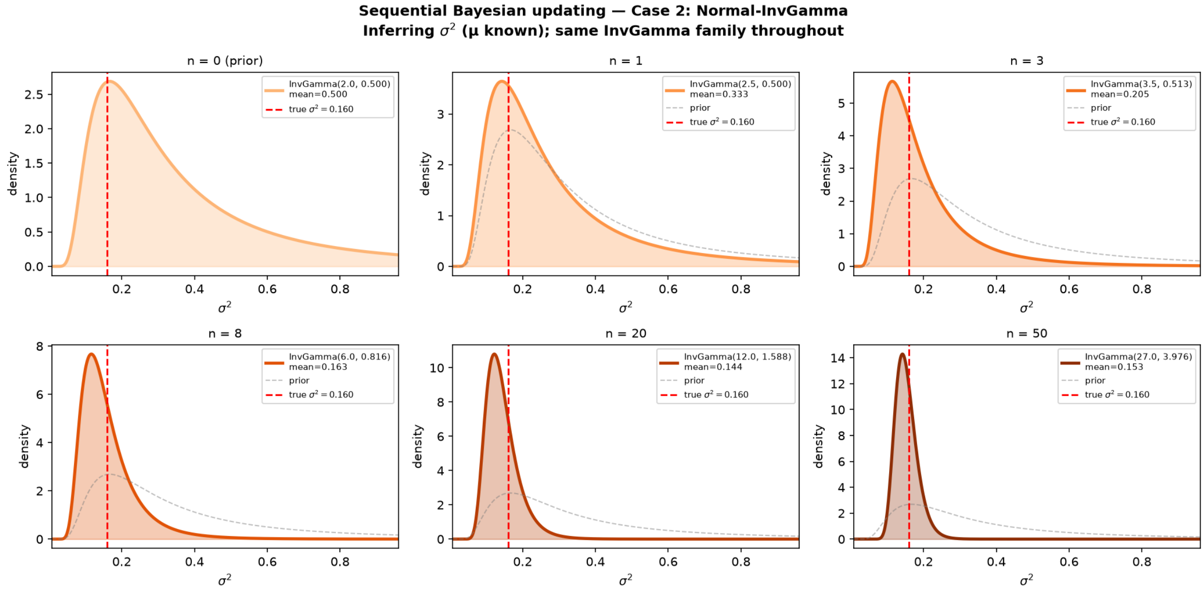

Figure 6: Sequential Normal-InvGamma updating at $n = 0, 1, 3, 8, 20, 50$. The InvGamma posterior concentrates near $\sigma^2_{\text{true}} = 0.16$ (red dashed). Throughout, the posterior remains an InvGamma distribution.

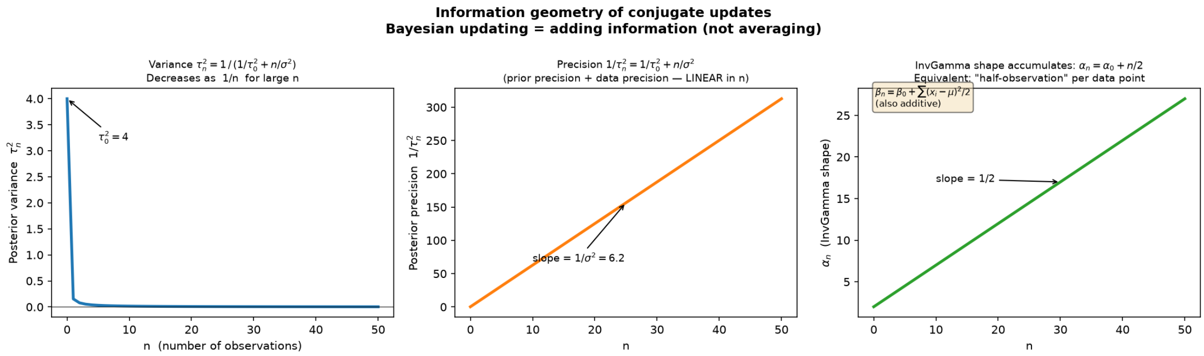

Figure 7: Information geometry of conjugate updates. *Left*: posterior variance τ_n^2 decreases hyperbolically in n. *Centre*: posterior precision $1/\tau_n^2$ is linear in n (information adds). *Right*: InvGamma shape parameter $\alpha_n = \alpha_0 + n/2$ is also linear ("half-observation" accumulation).

6.3 Case 3: Normal-Normal-InvGamma (joint inference of μ and σ^2)

Model. The joint Normal-InvGamma (NIG) prior factorises as

$$p(\mu, \sigma^2) = \mathcal{N}\left(\mu \,\middle|\, \mu_0, \frac{\sigma^2}{\kappa_0}\right) \cdot \text{InvGamma}(\sigma^2 \mid \alpha_0, \beta_0), \tag{22}$$

where κ_0 sets the strength of the prior on μ relative to the data noise.

Derivation. Given data $\{x_i\}_{i=1}^n$ with sample mean $\bar{x}$ and within-sample sum of squares $S = \sum(x_i - \bar{x})^2$, the log-posterior is

$$\begin{aligned}\log p(\mu, \sigma^2 \,|\, \mathbf{x}) &\propto \log p(\mathbf{x} \,|\, \mu, \sigma^2) + \log p(\mu \,|\, \sigma^2) + \log p(\sigma^2) \\ &= -\frac{n+1}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \left[\sum (x_i - \mu)^2 + \kappa_0 (\mu - \mu_0)^2 \right] - (\alpha_0 + 1) \log \sigma^2 - \frac{\beta_0}{\sigma^2}.\end{aligned} \tag{23}$$

Completing the square in μ (treating σ^2 as fixed):

$$\sum (x_i - \mu)^2 + \kappa_0 (\mu - \mu_0)^2 = (\kappa_0 + n)(\mu - \mu_n)^2 + S + \frac{\kappa_0 n (\bar{x} - \mu_0)^2}{\kappa_0 + n},$$

where $\mu_n = (\kappa_0 \mu_0 + n\bar{x})/(\kappa_0 + n)$. Substituting back:

Normal-InvGamma joint update

Posterior: $(\mu, \sigma^2) \,|\, \mathbf{x} \sim \mathrm{NIG}(\mu_n, \kappa_n, \alpha_n, \beta_n)$ with

$$\kappa_n = \kappa_0 + n \tag{24}$$

$$\mu_n = \frac{\kappa_0 \mu_0 + n\bar{x}}{\kappa_n} \tag{25}$$

$$\alpha_n = \alpha_0 + \frac{n}{2} \tag{26}$$

$$\beta_n = \beta_0 + \frac{S}{2} + \frac{\kappa_0 n (\bar{x} - \mu_0)^2}{2\kappa_n} \tag{27}$$

Marginal posteriors. Integrating out σ^2 gives the marginal for μ:

$$\mu \,|\, \mathbf{x} \;\sim\; t_{2\alpha_n}\left(\mu_n, \; \sqrt{\frac{\beta_n (\kappa_n + 1)}{\alpha_n \kappa_n}} \right),$$

a Student-t with $2\alpha_n$ degrees of freedom, heavier-tailed than a Gaussian. The marginal for σ^2 is simply $\mathrm{InvGamma}(\alpha_n, \beta_n)$.

Pseudo-data interpretation. κ_0 imaginary observations with mean μ_0 inform μ; $2\alpha_0$ imaginary observations with half-sum-of-squares β_0 inform σ^2. The real data updates both counters simultaneously.

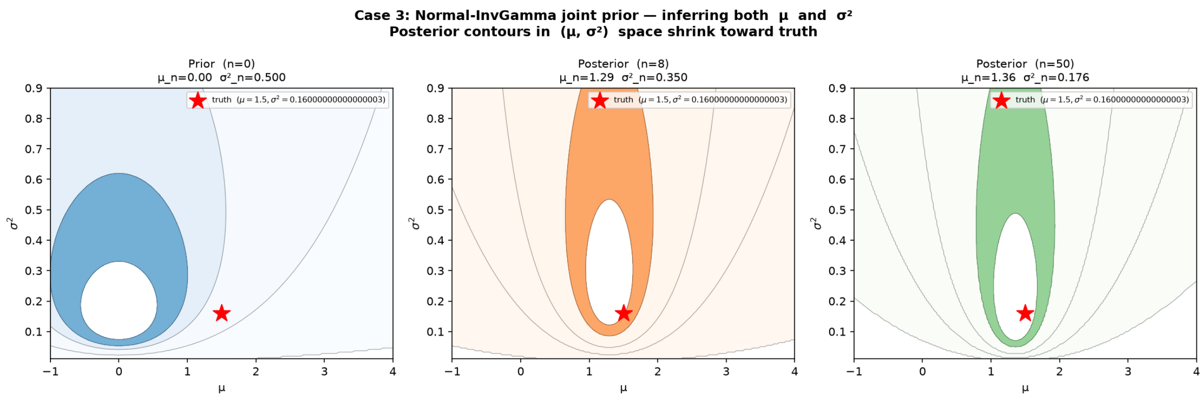

Figure 8: Joint NIG posterior contours in (μ, σ^2) space. *Left*: the prior spans a wide region. *Centre*: after $n = 8$ observations, the contours begin to concentrate. *Right*: after $n = 50$, the posterior is a tight ellipse around the true values (red star).

6.4 Connection to the Darcy Gibbs step

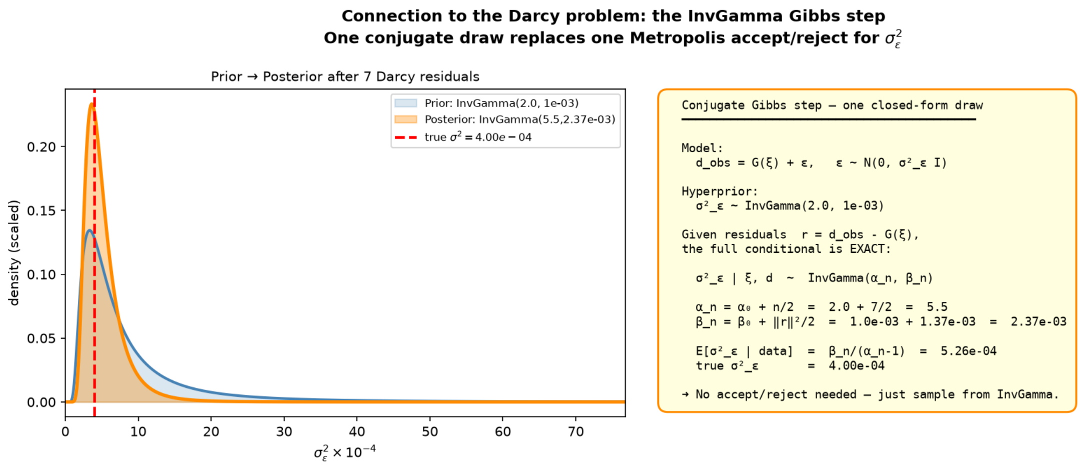

Figure 9: The InvGamma conjugate update applied to Darcy residuals. *Left*: the prior InvGamma is flat over the relevant range; after conditioning on 7 sensor residuals the posterior concentrates near the true σ_ε^2. *Right*: the closed-form update formula as used in the Gibbs step of the pCN sampler.

Table 1 summarises the three conjugate pairs.

Table 1: Conjugate pairs for the Gaussian likelihood.

Unknown	Prior family	Posterior family	Update rule
μ (σ known)	Normal	Normal	$\tau_n^{-2} = \tau_0^{-2} + n\sigma^{-2}$
σ^2 (μ known)	InvGamma	InvGamma	$\alpha_n = \alpha_0 + n/2,\ \beta_n = \beta_0 + \mathrm{SS}/2$
(μ, σ^2) joint	Normal-InvGamma	Normal-InvGamma	Eqs. (22)–(25)

7 The pCN-Gibbs MCMC Sampler

7.1 Why not random-walk Metropolis-Hastings?

A naive random-walk MH proposal $\xi^* = \xi + \varepsilon w$, $w \sim \mathcal{N}(0, I)$ suffers from the *curse of dimensionality*: the optimal step size $\varepsilon \sim M^{-1/2}$ shrinks as the number of modes M grows, causing the acceptance rate to collapse.

7.2 The pCN proposal

The preconditioned Crank-Nicolson (pCN) proposal is designed for targets of the form $p(\xi \mid \mathbf{d}) \propto p(\mathbf{d} \mid \xi)\, \mathcal{N}(\xi; 0, I)$:

$$\xi^* = \sqrt{1 - \beta^2}\, \xi + \beta\, w, \qquad w \sim \mathcal{N}(0, I_M), \quad \beta \in (0, 1). \tag{28}$$

Prior invariance. If $\xi \sim \mathcal{N}(0, I)$ then $\xi^* \sim \mathcal{N}(0, I)$: the proposal is *reversible with respect to the prior*. The Metropolis acceptance ratio therefore only involves the likelihood ratio:

$$\alpha(\xi, \xi^*) = \min\left(1,\ \frac{p(\mathbf{d}_{\mathrm{obs}} \mid \xi^*, \sigma_\varepsilon^2)}{p(\mathbf{d}_{\mathrm{obs}} \mid \xi, \sigma_\varepsilon^2)}\right) = \min\left(1,\ \exp\left[\frac{\|\mathbf{r}\|^2 - \|\mathbf{r}^*\|^2}{2\sigma_\varepsilon^2}\right]\right), \tag{29}$$

where $\mathbf{r} = \mathbf{d}_{\mathrm{obs}} - G(\xi)$ and $\mathbf{r}^* = \mathbf{d}_{\mathrm{obs}} - G(\xi^*)$.

Proof that prior and proposal ratios cancel. We verify the two claims in turn.

Claim 1: $\xi^* \sim \mathcal{N}(\mathbf{0}, I_M)$ whenever $\xi \sim \mathcal{N}(\mathbf{0}, I_M)$. Since ξ and w are independent standard Gaussians and ξ^* is an affine combination:

$$\begin{aligned}\mathbb{E}[\xi^*] &= \sqrt{1-\beta^2}\,\mathbb{E}[\xi] + \beta\,\mathbb{E}[w] = \mathbf{0},\\ \mathrm{Cov}[\xi^*] &= (1-\beta^2)\,I_M + \beta^2\,I_M = I_M.\end{aligned}$$

Hence $\xi^* \sim \mathcal{N}(\mathbf{0}, I_M)$, identical to the prior on ξ.

Claim 2: the joint density $p(\xi)\,q(\xi^*|\xi)$ is symmetric, so prior and proposal ratios cancel in the MH ratio. The pCN proposal kernel is $q(\xi^*|\xi) = \mathcal{N}(\xi^*; \sqrt{1-\beta^2}\,\xi,\, \beta^2 I)$. Expanding the squared norm:

$$\|\xi^* - \sqrt{1-\beta^2}\,\xi\|^2 = \|\xi^*\|^2 - 2\sqrt{1-\beta^2}\,\xi^\top\xi^* + (1-\beta^2)\|\xi\|^2,$$

so the full exponent of $p(\xi)\,q(\xi^*|\xi)$ is

$$-\tfrac{1}{2}\|\xi\|^2 - \frac{\|\xi^*\|^2 - 2\sqrt{1-\beta^2}\,\xi^\top\xi^* + (1-\beta^2)\|\xi\|^2}{2\beta^2} = -\frac{\|\xi\|^2 + \|\xi^*\|^2 - 2\sqrt{1-\beta^2}\,\xi^\top\xi^*}{2\beta^2}.$$

This expression is *symmetric* in (ξ, ξ^*), so $p(\xi)\,q(\xi^*|\xi) = p(\xi^*)\,q(\xi|\xi^*)$.

In the generic MH ratio:

$$A = \frac{p(\mathbf{d}|\xi^*)\cdot \overbrace{p(\xi^*)\,q(\xi|\xi^*)}^{=\,p(\xi)\,q(\xi^*|\xi)}}{p(\mathbf{d}|\xi)\cdot p(\xi)\,q(\xi^*|\xi)},$$

the prior–proposal product is identical in numerator and denominator and cancels, yielding $A = p(\mathbf{d}|\xi^*)/p(\mathbf{d}|\xi)$, which is Eq. (27). □

Dimension-robustness. The acceptance rate depends only on the likelihood increment, not on M. The same β works regardless of the number of KL modes.

7.3 Full sampler

Algorithm 1: pCN-Gibbs sampler for permeability inversion

Input: Data $\mathbf{d}_{\text{obs}}$, forward model G, KL object $\mathcal{K}$, step size β, hyperparameters (α_0, β_0), burn-in N_b, samples N_s, thinning Δ

Output: Posterior samples $\{\xi^{(t)}, \sigma_\varepsilon^{2(t)}\}$

Initialise $\xi^{(0)} \leftarrow \mathbf{0}$,;
$\sigma_\varepsilon^{2(0)} \leftarrow 10^{-3}$,;
$\mathbf{d}_{\text{cur}} \leftarrow G(\xi^{(0)})$;
for $t = 1$ **to** $N_b + N_s$ **do**
  Step 1 (pCN proposal for ξ);
  Draw $w \sim \mathcal{N}(0, I_M)$;
  Set $\xi^* \leftarrow \sqrt{1-\beta^2}\,\xi^{(t-1)} + \beta\, w$;
  Evaluate $\mathbf{d}^* \leftarrow G(\xi^*)$;
  Compute log-acceptance $\Delta\ell = \log p(\mathbf{d}_{\text{obs}} \,|\, \xi^*, \sigma^2) - \log p(\mathbf{d}_{\text{obs}} \,|\, \xi^{(t-1)}, \sigma^2)$;
  Draw $u \sim \text{Unif}(0, 1)$;
  if $\log u < \Delta\ell$ **then**
    $\xi^{(t)} \leftarrow \xi^*$;;
    $\mathbf{d}_{\text{cur}} \leftarrow \mathbf{d}^*$
  else
    $\xi^{(t)} \leftarrow \xi^{(t-1)}$
  end
  Step 2 (Gibbs draw for σ_ε^2);
  Set $\mathbf{r} \leftarrow \mathbf{d}_{\text{obs}} - \mathbf{d}_{\text{cur}}$;
  Draw $\sigma_\varepsilon^{2(t)} \sim \text{InvGamma}\big(\alpha_0 + m/2,\ \beta_0 + \|\mathbf{r}\|^2/2\big)$; `// Exact conjugate draw -- no MH`
  if $t > N_b$ ***and*** $(t - N_b) \bmod \Delta = 0$ **then**
    Store $(\xi^{(t)},\ \sigma_\varepsilon^{2(t)})$
  end
end

7.4 MCMC diagnostics

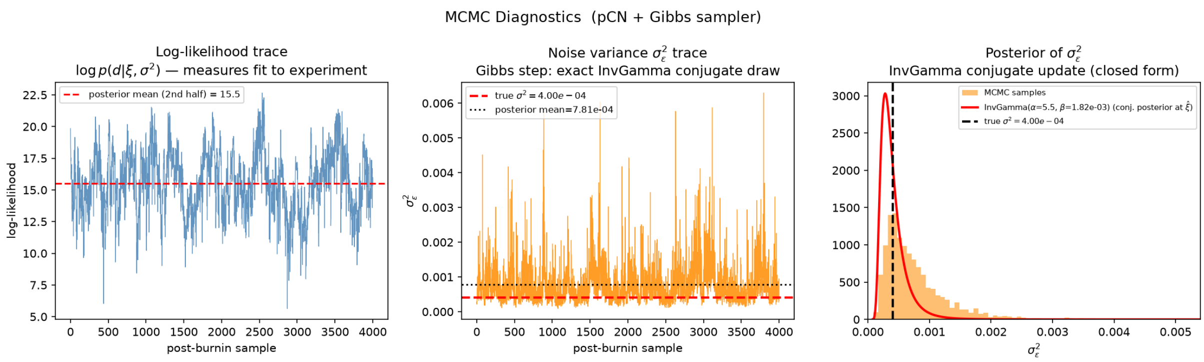

Figure 10: pCN-Gibbs MCMC diagnostics. *Left*: log-likelihood trace shows rapid mixing after the burn-in period. *Centre*: noise variance σ_ε^2 trace mixes around the true value (red dashed). *Right*: posterior histogram of σ_ε^2 overlaid with the analytical InvGamma posterior at the posterior-mean ξ, demonstrating that the Gibbs step is correctly implemented.

8 Posterior Results

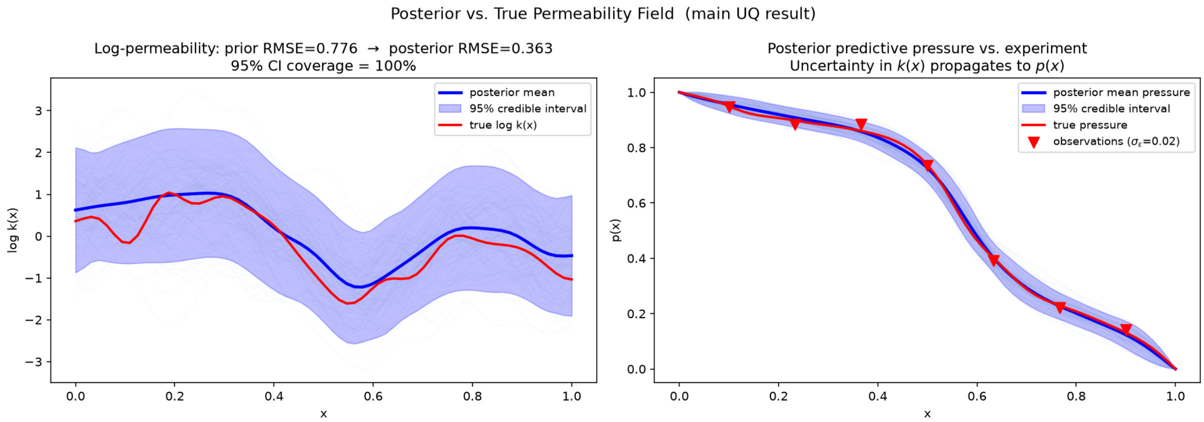

Figure 11: Posterior vs. true permeability field. *Left*: log-permeability posterior mean (blue), 95% pointwise credible interval (shaded), and true field (red). The prior RMSE (mean field vs. truth) of 0.78 is reduced to 0.36 in the posterior. Coverage of the 95% CI is $\approx$ 100% for this case, indicating slight over-coverage from the Gibbs-inferred noise variance. *Right*: posterior predictive pressure compared to the noisy observations. The uncertainty in $k(x)$ propagates to uncertainty in $p(x)$.

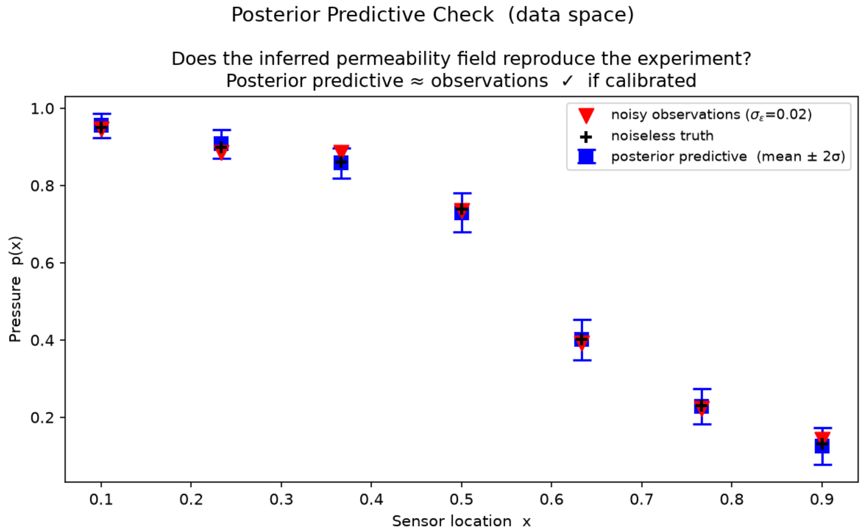

Figure 12: Posterior predictive check in data space. Blue squares show the posterior predictive mean $\pm 2\sigma$; red triangles are the noisy observations. The posterior predictive brackets the observations, confirming that the inferred field is consistent with the experiment.

8.1 Credible intervals versus confidence intervals

A 95% *credible interval* $[a, b]$ satisfies $P(k(x) \in [a, b] \,|\, \mathbf{d}_{\text{obs}}) = 0.95$—a direct probability statement about the unknown quantity given the data. This is distinct from a frequentist 95% confidence interval, which would state that 95% of *repeated experiments* would produce intervals containing the true value. In the UQ context, the Bayesian credible interval is the natural quantity: we have *one* experiment and wish to quantify uncertainty given that experiment.

9 Conclusions

We have traced the full chain from textbook probability distributions to a working posterior sampler for 1D single-phase Darcy permeability inversion:

1. The **Normal distribution** enters as the Gaussian likelihood (measurement error model), the GRF prior on $\log k(x)$, and the KL coefficients.
2. The **Inverse-Gamma** enters as the conjugate hyperprior on σ_ε^2, enabling an exact Gibbs update that replaces a Metropolis step.
3. The **Gamma** is the dual form (prior on precision), equally valid and leading to identical posterior calculations.
4. The **KL expansion** compresses the infinite-dimensional field inference into a finite-dimensional problem in $\mathbb{R}^M$, with the prior $\xi \sim \mathcal{N}(0, I)$ enabling the dimension-robust pCN proposal.
5. The **pCN-Gibbs** sampler alternates between a prior-reversible Metropolis step for ξ and an exact conjugate draw for σ_ε^2, producing samples from the full joint posterior.

The posterior log-k RMSE (0.36) is less than half the prior RMSE (0.78), and the 95% credible intervals comfortably cover the true field—concrete evidence that the 7 sensor readings carry meaningful information about the permeability structure.

A Complete Derivation of the NIG Joint Update

Starting from the log-posterior:

$$\log p(\mu, \sigma^2 \,|\, \mathbf{x}) \propto -\frac{n+1}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\Big[\sum (x_i - \mu)^2 + \kappa_0(\mu - \mu_0)^2 + 2\beta_0\Big].$$

Define $\kappa_n = \kappa_0 + n$ and $\mu_n = (\kappa_0\mu_0 + n\bar{x})/\kappa_n$. Expanding the quadratic in μ:

$$\begin{aligned}
\sum (x_i - \mu)^2 + \kappa_0(\mu - \mu_0)^2 &= \sum x_i^2 - 2n\bar{x}\mu + n\mu^2 + \kappa_0\mu^2 - 2\kappa_0\mu_0\mu + \kappa_0\mu_0^2 \\
&= \kappa_n\mu^2 - 2\kappa_n\mu_n\mu + (n\bar{x}^2 + \kappa_0\mu_0^2 - \kappa_n\mu_n^2) + S \\
&= \kappa_n(\mu - \mu_n)^2 + S + \frac{\kappa_0 n(\bar{x} - \mu_0)^2}{\kappa_n},
\end{aligned}$$

where $S = \sum (x_i - \bar{x})^2$ and the last equality uses $n\bar{x}^2 + \kappa_0\mu_0^2 - \kappa_n\mu_n^2 = \kappa_0 n(\bar{x} - \mu_0)^2/\kappa_n$ (verified by direct expansion).

Substituting:

$$\log p(\mu, \sigma^2 \,|\, \mathbf{x}) \propto -\frac{\kappa_n(\mu - \mu_n)^2}{2\sigma^2} - \Big(\alpha_0 + \frac{n}{2} + 1\Big)\log\sigma^2 - \frac{\beta_0 + S/2 + \kappa_0 n(\bar{x} - \mu_0)^2/(2\kappa_n)}{\sigma^2}.$$

This is the log of $\mathcal{N}(\mu \,|\, \mu_n, \sigma^2/\kappa_n)\cdot\mathrm{InvGamma}(\sigma^2 \,|\, \alpha_n, \beta_n)$ with α_n, β_n as given in Eqs. (24)–(25). □

B The Gamma Distribution as Conjugate Prior for Precision: Full Algebraic Derivation

This appendix derives in full why $\mathrm{Gamma}(\alpha_0, \beta_0)$ placed on the Gaussian *precision* $\tau = 1/\sigma^2$ produces a Gamma posterior. Every algebraic step is explicit.

B.1 Setup

Model.

$$x_i \mid \tau \overset{\text{i.i.d.}}{\sim} \mathcal{N}\!\left(\mu, \frac{1}{\tau}\right), \quad i = 1, \ldots, n, \qquad \mu \text{ known}, \quad \tau > 0 \text{ unknown}.$$

Prior.

$$\tau \sim \text{Gamma}(\alpha_0, \beta_0) \qquad \Longrightarrow \qquad p(\tau; \alpha_0, \beta_0) = \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)}\, \tau^{\alpha_0 - 1}\, e^{-\beta_0 \tau}, \quad \tau > 0.$$

Here β_0 is the *rate* (inverse scale), so $\mathbb{E}[\tau] = \alpha_0/\beta_0$ and $\text{Mode}[\tau] = (\alpha_0 - 1)/\beta_0$ for $\alpha_0 > 1$.

B.2 Gaussian density rewritten in precision form

Starting from $\mathcal{N}(x; \mu, \sigma^2)$ and substituting $\tau = 1/\sigma^2$:

$$\mathcal{N}\big(x; \mu, \tfrac{1}{\tau}\big) = \frac{1}{\sqrt{2\pi/\tau}} \exp\!\left(-\frac{(x-\mu)^2}{2/\tau}\right) = \sqrt{\frac{\tau}{2\pi}}\, \exp\!\left(-\frac{\tau(x-\mu)^2}{2}\right). \tag{30}$$

The variance σ^2 in the denominator has become the precision τ in the numerator of the exponent. This is the key structural fact: the likelihood is **linear in τ in the exponent**.

B.3 The likelihood

$$\begin{aligned}
p(\mathbf{x} \mid \tau) &= \prod_{i=1}^{n} \sqrt{\frac{\tau}{2\pi}}\, \exp\!\left(-\frac{\tau(x_i - \mu)^2}{2}\right) \\
&= (2\pi)^{-n/2}\, \tau^{n/2}\, \exp\!\left(-\frac{\tau}{2} \sum_{i=1}^{n} (x_i - \mu)^2\right) \\
&= (2\pi)^{-n/2}\, \tau^{n/2}\, \exp\!\left(-\frac{\text{SS}}{2}\, \tau\right),
\end{aligned} \tag{31}$$

where $\text{SS} = \sum_{i=1}^{n} (x_i - \mu)^2$ is the sum of squared deviations from the known mean.

The τ-dependent part of (29) is

$$\tau^{n/2}\, e^{-(\text{SS}/2)\,\tau}.$$

Compare this with the Gamma kernel $\tau^{\alpha_0 - 1}\, e^{-\beta_0 \tau}$: both are a power of τ times $\exp(-\text{const} \cdot \tau)$.

B.4 Posterior derivation (kernel recognition)

Applying Bayes' theorem and dropping all τ-free constants:

$$\begin{aligned}
p(\tau \mid \mathbf{x}) &\propto p(\mathbf{x} \mid \tau) \cdot p(\tau) \\
&\propto \left[\tau^{n/2}\, e^{-(\text{SS}/2)\tau}\right] \cdot \left[\tau^{\alpha_0 - 1}\, e^{-\beta_0 \tau}\right] \\
&= \tau^{\,n/2 \,+\, \alpha_0 \,-\, 1} \cdot \exp\!\Big(-\underbrace{(\beta_0 + \text{SS}/2)}_{\beta_n}\, \tau\Big) \\
&= \tau^{\underbrace{(\alpha_0 + n/2)}_{\alpha_n} - 1} \cdot e^{-\beta_n \tau}.
\end{aligned} \tag{32}$$

Expression (30) is the *unnormalised kernel* of $\text{Gamma}(\alpha_n, \beta_n)$. Since the Gamma normalising constant $\beta_n^{\alpha_n}/\Gamma(\alpha_n)$ is uniquely determined by (α_n, β_n), we conclude immediately:

Gamma-precision conjugate update

$$\tau \,|\, \mathbf{x}_1, \dots, \mathbf{x}_n \;\sim\; \mathrm{Gamma}(\alpha_n, \beta_n)$$

$$\alpha_n = \alpha_0 + \frac{n}{2} \qquad \text{(shape: } \tfrac{1}{2} \text{ added per observation)}$$

$$\beta_n = \beta_0 + \frac{\mathrm{SS}}{2} = \beta_0 + \frac{1}{2}\sum_{i=1}^{n}(x_i - \mu)^2 \qquad \text{(rate: squared deviations accumulate)}$$

B.5 Why the exponents collapse: exponential family structure

Both distributions belong to the *exponential family* in the natural parameter $\eta = -\tau$ (or equivalently, with sufficient statistic $T(\tau) = \tau$):

Distribution	Kernel	Natural param η	Suff. stat T
$\mathrm{Gamma}(\alpha_0, \beta_0)$ prior	$\tau^{\alpha_0 - 1} e^{-\beta_0 \tau}$	$-\beta_0$	τ
Gaussian likelihood	$\tau^{n/2} e^{-(\mathrm{SS}/2)\tau}$	$-\mathrm{SS}/2$	τ
Product (posterior kernel)	$\tau^{(\alpha_0 + n/2) - 1} e^{-(\beta_0 + \mathrm{SS}/2)\tau}$	$-(\beta_0 + \frac{\mathrm{SS}}{2})$	τ

Multiplying two exponentials with the same sufficient statistic simply *adds* the natural parameters:

$$e^{-\beta_0 \tau} \cdot e^{-(\mathrm{SS}/2)\tau} = e^{-(\beta_0 + \mathrm{SS}/2)\tau}.$$

This is the algebraic essence of conjugacy: the prior and likelihood share the same exponential structure in the parameter τ, so their product stays in the same exponential family.

B.6 Normalising constant and marginal likelihood

The full posterior normalisation gives the *marginal likelihood* $p(\mathbf{x})$ for free:

$$\begin{aligned} p(\mathbf{x}) &= \int_0^\infty p(\mathbf{x} \,|\, \tau)\, p(\tau)\, \mathrm{d}\tau \\ &= (2\pi)^{-n/2} \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)} \int_0^\infty \tau^{\alpha_n - 1} e^{-\beta_n \tau}\, \mathrm{d}\tau \\ &= (2\pi)^{-n/2} \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)} \cdot \frac{\Gamma(\alpha_n)}{\beta_n^{\alpha_n}}, \end{aligned} \tag{33}$$

using $\int_0^\infty \tau^{\alpha_n - 1} e^{-\beta_n \tau}\, \mathrm{d}\tau = \Gamma(\alpha_n)/\beta_n^{\alpha_n}$. Equation (31) is a product of Gamma and Student-t terms and is the quantity used in Bayesian model comparison.

B.7 Sequential consistency

Suppose we process the n observations one at a time. After datum x_1:

$$\alpha_1 = \alpha_0 + \tfrac{1}{2}, \qquad \beta_1 = \beta_0 + \tfrac{(x_1 - \mu)^2}{2}.$$

After datum x_2 (starting from $\mathrm{Gamma}(\alpha_1, \beta_1)$):

$$\alpha_2 = \alpha_1 + \tfrac{1}{2} = \alpha_0 + 1, \qquad \beta_2 = \beta_1 + \tfrac{(x_2 - \mu)^2}{2} = \beta_0 + \tfrac{(x_1 - \mu)^2 + (x_2 - \mu)^2}{2}.$$

After all n data:

$$\alpha_n = \alpha_0 + \frac{n}{2}, \qquad \beta_n = \beta_0 + \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{2} = \beta_0 + \frac{\mathrm{SS}}{2}.$$

This is identical to the batch update—the order of data presentation is irrelevant. This is sequentially consistent Bayesian updating.

B.8 Pseudo-data interpretation

The prior $\mathrm{Gamma}(\alpha_0, \beta_0)$ can be viewed as arising from $n_0 = 2\alpha_0$ *imaginary* (pseudo-) observations whose sum of squared deviations is $\mathrm{SS}_0 = 2\beta_0$. Then the posterior after n real observations is

$$\tau \,|\, \mathbf{x} \;\sim\; \mathrm{Gamma}\Big(\alpha_0 + \frac{n}{2},\; \beta_0 + \frac{\mathrm{SS}}{2}\Big) \equiv \mathrm{Gamma}\Big(\frac{n_0 + n}{2},\; \frac{\mathrm{SS}_0 + \mathrm{SS}}{2}\Big),$$

exactly as if we had $n_0 + n$ total observations with combined sum of squares $\mathrm{SS}_0 + \mathrm{SS}$. The prior encodes the same type of information as data; real data simply extends the pseudo-dataset.

B.9 The dual InvGamma via change of variables

Since $\tau > 0$, define $\sigma^2 = 1/\tau$. The Jacobian is $\mathrm{d}\tau/\mathrm{d}(\sigma^2) = -1/(\sigma^2)^2 = -\sigma^{-4}$, so

$$\begin{aligned}
p(\sigma^2) = p\Big(\tau = \frac{1}{\sigma^2}\Big) \cdot \left|\frac{\mathrm{d}\tau}{\mathrm{d}(\sigma^2)}\right| &= \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{\sigma^2}\right)^{\alpha-1} e^{-\beta/\sigma^2} \cdot \frac{1}{\sigma^4} \\
&= \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\sigma^2\right)^{-\alpha-1} e^{-\beta/\sigma^2}.
\end{aligned} \tag{34}$$

This is exactly $\mathrm{InvGamma}(\alpha, \beta)$. Therefore:

$$\tau \sim \mathrm{Gamma}(\alpha, \beta) \iff \sigma^2 = \frac{1}{\tau} \sim \mathrm{InvGamma}(\alpha, \beta).$$

The update formulas are identical in both parametrisations ($\alpha_n = \alpha_0 + n/2$, $\beta_n = \beta_0 + \mathrm{SS}/2$); only the axis label changes from τ to $\sigma^2 = 1/\tau$.

Parametrisation	Prior	Posterior
Precision $\tau = 1/\sigma^2$	$\mathrm{Gamma}(\alpha_0, \beta_0)$	$\mathrm{Gamma}(\alpha_0 + n/2,\; \beta_0 + \mathrm{SS}/2)$
Variance $\sigma^2 = 1/\tau$	$\mathrm{InvGamma}(\alpha_0, \beta_0)$	$\mathrm{InvGamma}(\alpha_0 + n/2,\; \beta_0 + \mathrm{SS}/2)$

B.10 Summary of the conjugacy argument

The proof reduces to one observation: the Gaussian likelihood $p(\mathbf{x}|\tau) \propto \tau^{n/2} e^{-(\mathrm{SS}/2)\tau}$ and the Gamma prior $p(\tau) \propto \tau^{\alpha_0 - 1} e^{-\beta_0 \tau}$ are both of the form $\tau^c \cdot e^{-d\tau}$. Multiplying two such expressions yields another expression of the same form with updated exponent c and rate d:

$$\underbrace{\tau^{\alpha_0-1}\, e^{-\beta_0\tau}}_{\text{prior}} \times \underbrace{\tau^{n/2}\, e^{-(\mathrm{SS}/2)\tau}}_{\text{likelihood}} \;=\; \tau^{(\alpha_0+n/2)-1}\, e^{-(\beta_0+\mathrm{SS}/2)\tau} \;\propto\; \underbrace{\mathrm{Gamma}\big(\alpha_0 + \tfrac{n}{2},\; \beta_0 + \tfrac{\mathrm{SS}}{2}\big)}_{\text{posterior}}.$$

The family is preserved because the product of two power-times-exponential functions is again a power-times-exponential function—and that is precisely the Gamma family.